Multivariate Calculus Lecture Notes
Section 14.6 – The Chain Rule

I. The chain rule for functions of two independent variables
   a. We've talked about the chain rule for multivariate functions stuffed into single-variable functions (chain rule for gradients) and single-variable functions stuffed into multivariate functions (chain rule for paths), but what about multivariate functions stuffed into multivariate functions?
   b. Suppose we have a function $f(x, y, z)$, where $x = x(s, t)$, $y = y(s, t)$, and $z = z(s, t)$. How do we calculate $\dfrac{\partial f}{\partial s}$?
      i. If we're interested in $\dfrac{\partial f}{\partial s}$ at a point $P = (s_0, t_0)$ that means we're holding t constant at t_0 and looking at the resulting path as s_0 changes.
      ii. Thus we're considering the path $\mathbf{c}(s) = \langle x(s, t_0), y(s, t_0), z(s, t_0)\rangle$.
      iii. According to the chain rule for paths, we have
      $$\frac{\partial f}{\partial s} = \nabla f_{\mathbf{c}(s)} \cdot \mathbf{c}'(s) = \frac{\partial f}{\partial x}\frac{\partial x}{\partial s} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial s} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial s}.$$
      iv. Similarly, if $\mathbf{d}(t)$ is the path in the direction of increasing t through the point (s_0, t_0),
      $$\frac{\partial f}{\partial t} = \nabla f_{\mathbf{d}(t)} \cdot \mathbf{d}'(t) = \frac{\partial f}{\partial x}\frac{\partial x}{\partial t} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial t} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial t}.$$
   c. A similar idea extends to functions of three or more independent variables.
   d. Example (#3): Find $\dfrac{\partial f}{\partial s}$ and $\dfrac{\partial f}{\partial r}$ if $f(x, y, z) = xy + z^2$, $x = s^2$, $y = 2rs$, and $z = r^2$.
      i. In both cases we'll need ∇f, so let's calculate that first: $\nabla f = \langle y, x, 2z\rangle$.
      ii. We'll also need all of the partial derivatives:
      $$\frac{\partial x}{\partial s} = 2s \qquad \frac{\partial x}{\partial r} = 0$$
      $$\frac{\partial y}{\partial s} = 2r \qquad \frac{\partial y}{\partial r} = 2s$$
      $$\frac{\partial z}{\partial s} = 0 \qquad \frac{\partial z}{\partial r} = 2r$$
      iii. Putting this all together, we have
      $$\frac{\partial f}{\partial s} = 2sy + 2rx = 2s(2rs) + 2r(s^2) = 6rs^2$$
      $$\frac{\partial f}{\partial r} = 2s(x) + 2r(2z) = 2s(s^2) + 4r(r^2) = 2s^3 + 4r^3.$$

II. Implicit partial differentiation
   a. The chain rule leads to a shortcut formula for implicit partial differentiation (which we haven't really talked about yet).
      i. Suppose
         1. The function $F(x, y, z)$ is differentiable.
         2. The equation $F(x, y, z) = 0$ defines z implicitly as a differentiable function of x and y.

3. Suppose we want to find $\dfrac{\partial z}{\partial x}$ and $\dfrac{\partial z}{\partial y}$.

ii. Then, by the chain rule, we have

$$0 = \frac{\partial F}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial F}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial F}{\partial z}\frac{\partial z}{\partial x}$$

$$\Longrightarrow 0 = F_x(1) + F_y(0) + F_z\frac{\partial z}{\partial x}$$

$$\Longrightarrow \frac{\partial z}{\partial x} = -\frac{F_x}{F_z}.$$

iii. Also, $\dfrac{\partial z}{\partial y} = -\dfrac{F_y}{F_z}$.

iv. This is a lot faster than doing implicit differentiation the usual way.

b. Find the first partial derivatives of z, where $x \ln y + y^2 z + z^2 = 8$.

i. Partials of $F(x, y, z) = x \ln y + y^2 z + z^2 - 8$

1. $F_x = \ln y$
2. $F_y = x/y + 2yz$
3. $F_z = y^2 + 2z$.

ii. Thus

$$\frac{\partial z}{\partial x} = \frac{-\ln y}{y^2 + 2z},$$

$$\frac{\partial z}{\partial y} = \frac{-x/y - 2yz}{y^2 + 2z} = \frac{-x - 2y^2 z}{y^3 + 2yz}.$$

Homework: Section 14.6 #4, 10, 11, 18, 23, 27, 35.